Differential Expression with limma-voom

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Introduction

limma-voom is one of the three dominant differential-expression methods for RNA-seq (alongside DESeq2 and edgeR). The voom step transforms raw counts onto the log-CPM scale and computes per-observation precision weights that account for the mean-variance trend characteristic of count data. The result is suitable input to limma's full linear-modelling framework, which adds the empirical-Bayes shrinkage of variance estimates that drives limma's power on small samples and complex designs.

Prerequisites

A working understanding of linear models, RNA-seq normalisation (TMM via edgeR), the mean-variance relationship of count data, and the empirical-Bayes shrinkage approach.

Theory

`voom()` transforms raw counts via $\log_2(\mathrm{CPM} + 0.5)$ and fits a smooth mean-variance trend across genes. The fitted trend produces a precision weight for each observation; downstream `lmFit()` regression uses these weights so that low-expression genes (with high relative variance) contribute less to the joint estimation. The empirical-Bayes step (`eBayes()`) shrinks per-gene variance estimates toward the global trend, yielding moderated t-statistics that have better small-sample properties than ordinary t-tests.

The full limma-voom pipeline supports arbitrarily complex designs (interactions, blocking, batch adjustment) through standard `model.matrix` notation, and the contrast machinery handles multi-condition comparisons cleanly via `contrasts.fit()`.

Assumptions

Counts have been TMM-normalised (via `calcNormFactors` from edgeR); the chosen design matrix correctly encodes the biological and technical factors; the mean-variance trend is approximately monotone (true for typical bulk RNA-seq).

R Implementation

```r
library(limma); library(edgeR)

set.seed(2026)
counts <- matrix(rpois(20000 * 12, lambda = 30), 20000, 12)
group <- factor(rep(c("A", "B"), each = 6))

dge <- DGEList(counts = counts); dge <- calcNormFactors(dge)
design <- model.matrix(~ group)

v <- voom(dge, design, plot = FALSE)
fit <- lmFit(v, design)
fit <- eBayes(fit)
```

```r
topTable(fit, coef = 2, n = 10)
summary(decideTests(fit))
```

Output & Results

`topTable()` returns the top differentially-expressed genes with $\log_2$ fold change, average expression, moderated t-statistic, raw p-value, and Benjamini-Hochberg adjusted p-value. `decideTests()` summarises significance calls per contrast, useful for cross-method comparison.

Interpretation

A reporting sentence: "limma-voom identified 380 genes at FDR < 0.05 between conditions A and B ($n = 6$ per group), with mean fold change among significant genes of 1.7 (range 0.4 to 3.8); the corresponding edgeR analysis on the same data identified 342 genes, with 320 genes (84 %) shared between methods, indicating substantial concordance." Always state the FDR threshold and the design matrix structure.

Practical Tips

- limma-voom is competitive with edgeR's `glmQLFit` and DESeq2's negative-binomial Wald test; choose based on workflow conventions and personal preference rather than expected accuracy gains.
- `voomWithQualityWeights()` additionally down-weights noisy samples, useful when one or two samples have visibly poor QC; combine the two weight types when sample-level variability is substantial.
- For complex designs (interactions, multiple blocking factors), the standard `model.matrix` and `contrasts.fit()` machinery applies; limma's design language is among the most flexible.
- Voom is well-suited to scRNA-seq pseudobulk: aggregate cells per cluster into pseudo-bulk counts, then run voom on the resulting matrix.
- For very few samples per group ($n < 3$), variance estimates are unstable; consider DESeq2 with its variance-stabilising shrinkage, or rely heavily on the empirical Bayes step in limma.
- Pair the analysis with `voomQCPlot` or `meanvariancetrend()` diagnostic plots to confirm the variance trend is well-fitted.

R Packages Used

`limma` for `voom()`, `lmFit()`, `eBayes()`, `topTable()`, and the broader linear-modelling framework; `edgeR` for `DGEList`, `calcNormFactors` (TMM normalisation); `Glimma` for interactive MA and volcano plots; `EnhancedVolcano` for publication-ready volcano plots from limma output.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- RNA-seq with DESeq2 / edgeR
- Enrichment analysis (GSEA / ORA)
- scRNA-seq with Seurat